# PLA/PBAT/MCC/HA Composite Optimized for use as Biodegradable Mulch

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**Abstract**

Biodegradable polymers are an emerging solution for the tons of plastic waste produced by the global agriculture industry annually. Not only that, fertilizers also contribute to plastic waste. Biodegradable mulch films, intended to be tilled into the soil after its use during a growing season, can serve as a strategy to both fertilize soil and conserve resources as a mulch. The study herein explores poly(lactic acid), poly(butylene adipate-*co*-terephthalate), microcrystalline cellulose, and hydroxyapatite.

**Keywords:** Biodegradable polymers, PLA, PBAT, mulch, biodegradable mulch

## 1 Introduction

Mulch is a critical part of modern agriculture that conserves water, promotes seed germination and root health, prevents pests and weeds, and maintains temperature [1]. Simultaneously, the conventional material for mulch, low density poly(ethylene) (LDPE) is a significant contributor to microplastic pollution [2]. Microplastics found in agriculture soil, from sources such as mulches or municipal wastewater diverted for crop irrigation [3], are known to be incorporated into our food chain via crop uptake [4]. The presence of microplastics in agriculture soil is therefore an emerging problem that directly contributes to the introduction of microplastics in the human body.

Biodegradable polymers are promising alternatives to LDPE with options already on the market [5]. Between LDPE and biodegradable films, there appears to be little to no difference in crop quality for short growing seasons (e.g. onions between June 22 and August 10, 49 days [5, 6]). However, with longer growing seasons, premature degradation of films result in non-suitable performances [7]. Current commercial options available lack sufficient durability for a full growing season in the presence of harsh abiotic stresses like solar irradiation, high temperatures, and precipitation [7, 8].

Not only do current BDMs lack sufficient durability as a film, they also do not mineralize within 2 years, as recommended by commercial guidelines [9]. While commercially compostable, degradability is slow in realistic end-user conditions [6, 10, 11]. Commercial PBAT and/or PLA-based BDMs achieve <30% biodegradation even after as much as 400 days in soil [12]. Degradation is environment-dependent [10, 11]. For example, a study by Sintim et al. 2020 [10] found that degradation (as quantified by surface area reduction) was significantly faster in compost (85-99% degradation after 18 weeks) than in soil (26-83% in soil after 36 months). Polymer decomposition is a complex phenomenon with multiple concurrent processes like oxidation, chain scission, hydrolysis, mechanical decomposition, biofilm formation, enzymatic digestion (i.e. laccases, esterases), and mineralization [8, 11, 13]. When above the ground, abiotic factors like sun irradiance and water are highly relevant. After incorporation into soil, where sunlight is significantly diminished, biodegradation takes over. Films should be as robust as possible during use on the soil, but then degrade as quickly as possible once incorporated into the soil after use. One strategy for overcoming slow degradation in soil is to blend hydrophobic polymer matrices with smaller, hydrophilic additives. For example, Kansara et al. [6] developed a multi-layer film with PBAT external layers and thermoplastic starch inner layers with enhanced biodegradability [6]. Microcrystalline cellulose (MCC) is a semi-crystalline form of cellulose where the amorphous regions are preferentially hydrolyzed with acid, leaving majority crystalline structure [14]. The crystalline structure is maintained by Van der Waals forces and hydrogen bonding between adjacent hydroxyl groups [15]. Hydroxyapatite (HA) is a promising material for use in agriculture. For example, Wang et al. [16] found that augmenting soil with HA and an HA-solubilizing bacterium, bacterium *Klebsiella sp. M2* (from radish rhizosphere), released $PO_4^{-3}$ and $Ca^{2+}$ ions that precipitated the water-insoluble compound $Cd_3(PO_4)_2$, thereby avoiding Cd uptake in lettuce crops [16]. Not only that, HA augmentation increased crop yield and $Ca^{2+}$ crop content, which was further enhanced in the presence of Klebsiella sp. M2 [16]. Yan et al. 2025 [17] had promising results with Cd remediation of rice paddy soils, where the amount of HA correlated with the bio available Cd content in the soil.

The work herein utilizes a design of experiment (DoE) approach in analyzing the effects of PLA, PBAT, MCC, and HA composition on mechanical and water absorption properties with the objective of developing a composite with enough strength, durability, and water absorption for application as BDM.

# 2 Methods

## 2.1 Materials

The materials used include Ingeo™ 4032D semi-crystalline high-heat film grade poly(lactic acid) (PLA) (NatureWorks, Minneapolis, MN, USA), Ecoflex® F blend C1200 poly(butylene adipate-*co*-terephthalate) (PBAT) (BASF, Ludwigshafen, Germany), MCC (Sigma-Aldrich Inc., St. Louis, MO, USA), and HA (synthetic reagent grade, Sigma-Aldrich Inc., St. Louis, MO, USA). All compositions are reported in mass percentages. Sample weights were measured via analytical scale (Fisher Scientific™ Education ALF104, Hampton, NH, USA) with ± 0.0001 g accuracy. Commercial LDPE (76.2 $\mu$m thick embossed red mulch plastic film) was purchased as a control from the seller, Adnee Store, on Amazon [18]. A reference commercial BDM (Biodegradable Mulch Film, Natural Weed Control for Gardens & Farms (48" × 50')) was purchased from the seller, The Gardener's Workshop Turning All Thumbs Green, on Amazon [19]. The commercial BDM was manufactured by Dubois Agrinovation under the trade name Bio360®, which is composed of Mater-Bi® (a product line of proprietary blends of biodegradable polymers by Italian company Novamont S.p.A.) and vegetable starch. Mater-Bi and Bio360® are EN 13432 and ASTM Standard D 6400-23 compliant according to specification sheets [20]. The commercial BDM had an average thickness of 18.3 $\mu$m (n=3). All materials and samples were stored under vacuum or nitrogen glove box (Vacuum Technology Inc., Oak Ridge, TN, USA) with oxygen and water vapor concentrations less than 300 and 0.1 ppm, respectively.

## 2.2 Design of Experiment (DoE)

A full factorial extreme vertices design of experiments (DoE) with one center point ($7^D$), eight vertices ($2^4 = 16$ runs), and three replicates for a total of $17 \times 3 = 51$ runs (Table 1) was developed using Minitab® Statistical Software (Version 22.4.0, Minitab LLC, State College, PA USA). Pure PLA ($18^C$), pure PBAT ($19^C$), pure MCC ($20^C$), and pure HA ($21^C$) were also used as controls. The factors

include PLA concentration (5-30%), PBAT concentration (70-95%), MCC concentration (0-10%), and HA concentration (0-10%). Control and validation runs were only plotted for reference. The data collected were either part of the DoE design space (runs 1–17), control samples (runs 18–21), or validation samples (22-28) (Table 1).

Elastic Net regression models were implemented in Python using the ElasticNetCV class from the scikit-learn library (Version 1.7.2, Anaconda Inc, Austin, Texas, USA). Second-order interaction terms (degree = 2, interaction only, no quadratic self-terms, zero intercept) were generated using polynomial feature expansion. Predictor variables were standardized prior to model fitting, and response variables were standardized using a transformed target approach. Hyperparameters ($\alpha$ and l1 ratio) were optimized via 5-fold cross-validation over a predefined grid ($\alpha \in [10^{-6}, 10^{-1}]$; l1 ratio $\in [0.5, 1]$) for each response. Models were trained in standardized space and coefficients were subsequently transformed back to original units for interpretation. The final parameters used can be found in Supporting Information. DoE data (runs $1^{D}$- $17^{D}$) were used for model training. Control and validation data were excluded from all model training.

**Table 1** Runs 1–17 were devised by Minitab® DoE software. Run $7^{D}$ is a center point while the others are vertices. Runs 18–21 are control samples that were not used in DoE regression.

| Run ID | PLA (wt%) | PBAT (wt%) | MCC (wt%) | HA (wt%) |
|---|---|---|---|---|
| $1^{D}$ | 30 | 70 | 0 | 0 |
| $2^{D}$ | 21.25 | 73.75 | 2.5 | 2.5 |
| $3^{D}$ | 20 | 70 | 10 | 0 |
| $4^{D}$ | 20 | 70 | 0 | 10 |
| $5^{D}$ | 16.25 | 73.75 | 7.5 | 2.5 |
| $6^{D}$ | 16.25 | 73.75 | 2.5 | 7.5 |
| $7^{D}$ | 12.5 | 77.5 | 5 | 5 |
| $8^{D}$ | 11.25 | 73.75 | 7.5 | 7.5 |
| $9^{D}$ | 10 | 70 | 10 | 10 |
| $10^{D}$ | 8.75 | 81.25 | 7.5 | 2.5 |
| $11^{D}$ | 8.75 | 86.25 | 2.5 | 2.5 |
| $12^{D}$ | 8.75 | 81.25 | 2.5 | 7.5 |
| $13^{D}$ | 8.75 | 76.25 | 7.5 | 7.5 |
| $14^{D}$ | 5 | 75 | 10 | 10 |
| $15^{D}$ | 5 | 95 | 0 | 0 |
| $16^{D}$ | 5 | 85 | 0 | 10 |
| $17^{D}$ | 5 | 85 | 10 | 0 |
| $18^{C}$ | 100 | 0 | 0 | 0 |
| $19^{C}$ | 0 | 100 | 0 | 0 |
| $20^{C}$ | 0 | 0 | 100 | 0 |
| $21^{C}$ | 0 | 0 | 0 | 100 |
| $22^{V}$ | 10.5 | 83.8 | 2.8 | 2.8 |
| $23^{V}$ | 10.3 | 81.1 | 2.9 | 5.7 |
| $24^{V}$ | 5 | 90 | 2.5 | 2.5 |
| $25^{V}$ | 8 | 88 | 2 | 2 |
| $26^{V}$ | 9 | 85 | 3 | 3 |
| $27^{V}$ | 10 | 85 | 2.5 | 2.5 |
| $28^{V}$ | 29 | 66 | 2.5 | 2.5 |

The superscripts $^{D}$, $^{C}$, and $^{V}$ represent the DoE, control, and validation datasets, respectively. Run $20^{V}$ is the optimized composition.

## 2.3 Film Preparation

Films were prepared using commercial feedstocks of polymer resin pellets (PBAT, PLA) and powder additives (MCC, HA). The particles were first suspended ("pre-mixed") in the polymer matrix by heating them inside the hot melt press without applying force at 170°C. This step was critical in reducing losses during loading the powders (MCC, HA) into the hot melt mixer, thus preserving the actual concentrations. Without this step, loading the powder into the mixer was not possible due to powder lifting while loading into the small mixing chamber. The "pre-mixed" blends/composites were then loaded into the hot melt mixer. For consistency, blends without powder additives (e.g. pure PLA) were also subjected to this "pre-mixing step" before melt mixing. A HaakeBuchler Rheocord

40 (Thermo Fisher Scientific, Waltham, MA, USA) double-arm mixing unit was used to mix batch sizes of 40 g with roller mixing blades at 180°C and 55 rpm until torque stabilized (<10 minutes for all blends). The melt-mixed blends/composites were subsequently loaded into a bronze cavity (13 cm diameter, 0.033 cm height) for hot melt pressing via hydraulic press (Carver Laboratory Press, Wabash, IN, USA) at 0.84 MPa and 170°C for 90 seconds. Films were cooled to room temperature using cold water within 5 minutes at a rate of approximately 1.5°C·s$^{-1}$. A 7326A (Mitutoyo, Kawasaki City, Kanagawa, Japan) dial thickness gauge (± 0.005 mm) or a 700-118-30 (Mitutoyo, Kawasaki City, Kanagawa, Japan) digital thickness gauge (± 0.001 mm) were used to measure film thickness prior to mechanical characterization.

## 2.4 Thermogravimetric Analysis (TGA)

Degradation parameters, moisture content, and char yield were evaluated using thermogravimetric analysis (TGA) on a Thermogravimetric Analyzer TGA 8000 (PerkinElmer Inc., Waltham, MA, USA). Degradation parameters include the temperatures at 10% mass loss (90% remaining mass, $T_{90}$), 15% mass loss (85% remaining mass, $T_{85}$), and 50% mass loss ($T_{50}$) where $T_{85}$ is considered the decomposition onset [21]. Samples were loaded into the platinum pan with an approximately 10 mg sample loading per run. The temperature program was performed under 20 mL·min$^{-1}$ nitrogen gas flow as follows: isotherm for 5 min at 30°C followed by heating from 30°C to 600°C at 10°C·min$^{-1}$.

## 2.5 Water Absorption

Water absorption was measured in triplicate using guidance from ASTM Standard D 570-22 [22]. Films were cut to 80 mm squares and submerged in deionized water equilibrated to atmospheric conditions inside glass or poly(styrene) Petri dishes. In cases where the films were prone to float to the top, an O ring was used to keep it submerged. 80 mm squares were selected because they contained sufficient mass to detect weight changes using our ALF104 (Fisher Scientific Education, Waltham, MA, USA) analytical scale with ± 0.0001 g accuracy. The pH of the water was 5.72 ± 0.06 (n=5) under atmospheric conditions, measured using Accumet AB250 (Fisher Brand, Waltham, MA, USA) pH meter. Water absorption was measured until there were two consecutive weight changes <5 mg, in accordance with ASTM Standard D 570-22 [22].

## 2.6 Fourier-Transform Infrared Spectroscopy (FTIR)

Fourier-Transform Infrared Spectroscopy (FTIR) was used to probe quantitative and qualitative signatures of degradation. FTIR traces were obtained using a Cary 620 (Agilent Technologies, Inc., Santa Clara, California, USA) via diamond Attenuated Total Reflectance (ATR). Scanning was performed using 32 scans, 2 cm$^{-1}$ resolution, and air background subtraction. An approach utilized by Graf et al. [11] was used, where the carbonyl index (CI) and hydroxyl index (HI) indicate the degree of degradation. CI and HI are calculated using Equation 1:

$$Index = \frac{Area_{\text{Functional Group}}}{Area_{\text{C-H}}}, \tag{1}$$

where the functional group is either the C=O or the OH group. For LDPE, the C=O wavenumber was taken between 1536-1928 cm$^{-1}$ while the C-H bonds were calculated between 1384-1472 cm$^{-1}$, as selected by Graf et al. [11]. For PLA/PBAT blends it was calculated using the C=O integral between 1552-1808 cm$^{-1}$ while the C-H bonds were calculated between 1336-1496 cm$^{-1}$, as selected by Graf et al. [11]. For all samples, the OH integral was taken over X-Y cm$^{-1}$ [REFERENCE].

## 2.7 Mechanical Testing

Mechanical testing was performed on an Instron® 5967 (Instron®, Norwood, MA, USA) or Instron® 68FM-300 (Instron®, Norwood, MA, USA) fitted with tapered grips. ASTM Standard D 638-22 Type V dog-bones were cut with ASTM-certified cutting dies (Fremont Cutting Dies, Inc., Fremont, OH, USA). Elongation at break ($\varepsilon_{\text{Break}}$) and strength at break ($\sigma_{\text{Break}}$) were characterized in triplicate for each sample from engineering stress-strain curves obtained at 50 mm·min$^{-1}$ testing speed and room temperature. Calculations and testing conditions followed guidance from ASTM Standard D 638-22 [23]. Specific equations are provided in Supporting Information. All dog-bone replicates were cut alongside the same direction within each film. Each composition was characterized pristine (as prepared), after UV aging, and after submersion in water for 80 days (saturated in water). Compact

letter display using Tukey Honestly Significant Difference (HSD) was used to display significant differences (alpha = 0.05) between a pristine sample, a sample exposed to UV aging, and a sample that was saturated in water for 80 days. Each composition is its own group – that is, significance was calculated between the means of a single composition pristine, UV-aged, and saturated with water (n=3). The letters displayed within a composition have no relation to those displayed for another composition and only reflect groups based on treatment (pristine, UV-aged, and saturated with water). Analyses were performed using Python (Version 3.10.9, Anaconda Inc, Austin, Texas, USA) SciPy library.

### 2.8 Ultraviolet (UV) Accelerated Aging

Ultraviolet aging experiments were conducted inside a laboratory fume hood equipped with air circulation to remove reactive gases, eight T12 flourescent tube (Q-Lab Corporation, Westlake, OH, USA) 340 nm light sources, and two IONBOARD S22 LED Grow Light Boards with Samsung LM301H Diodes (AC Infinity, City of Industry, California, USA). Accessories turned on and off in 8-hr on/4-hr off cycles, in accordance with ASTM Standard D 154-23 [24]. Temperature was recorded using a Controller 69 Pro (AC Infinity, City of Industry, California, USA) fitted with a temperature probe. A film of each sample was used and three dog bones were cut from each sample for mechanical analysis after aging. Cumulative UV irradiation, H ($J \cdot m^{-2}$), is calculated using irradiance, E ($W \cdot m^{-2}$), and irradiation time, t (s) (Equation 2):

$$H = E \cdot t \tag{2}$$

Irradiance was measured using a UV513AB (General Tools & Instruments, Secaucus, NJ, USA) 280-400 nm light meter with ± 4% resolution. All samples were maintained 20 cm from the light source, where irradiance was measured ten times and averaged before calculating the experiment duration. Exposure time was calculated to simulate 3 months of summer irradiance using Rutgers New Jersey Weather Network irradiance data [25]. Samples which were not exposed to accelerated aging are labeled "pristine" while samples subjected are denoted "aged." Film mass was recorded prior to exposure ($m_0$) and then after exposure ($m_t$) to calculate mass change ($\Delta m$) (Equation 3).

$$\Delta m = \frac{m_t - m_0}{m_0} \tag{3}$$

### 2.9 X-Ray Diffraction (XRD)

XRD was characterized using a PANalytical Empyrean (Malvern Panalytical Ltd., Almelo, The Netherlands) X-ray diffractometer with a 27 mm monocrystalline Si zero-background sample holder. The operating conditions involved 45 kV and 40 mA using Cu K$\alpha$ radiation ($\lambda$ = 1.5438 Å). Crystalline concentration (X) of each blend was calculated using the following equation (Equation 4:

$$X = \frac{I_{\text{crystalline}}}{I_{\text{amorphous}} + I_{\text{crystalline}}} \cdot 100 \tag{4}$$

where $I_{crystalline}$ and $I_{amorphous}$ are the areas under the crystalline and amorphous signals of the XRD diffractograms [26]. Note that the instrument background signal ("constant background signal") was subtracted from all diffractograms prior to calculation of X. The crystallize size (D) was calculated using the Scherrer equation below (Equation 5):

$$D = \frac{K \cdot \lambda}{\beta \cdot cos(\theta)} \tag{5}$$

where $\beta$ is the peak's full width at half maximum (FWHM) in radians; $\lambda$ = 0.15438 nm; $\theta$ is the Bragg angle (peak position) in radians; and K is the shape factor. K was assumed to be 0.9 for PLA, PBAT, and HA [27, 28], and 0.94 for MCC [29].

## 3 Results and Discussion

### 3.1 Thermal Properties

Thermal properties served as indicators of composition and provided insight into thermal stability.

The thermal stability of the PLA/PBAT-based systems was strongly influenced by matrix composition and filler incorporation (particularly HA) (Figure 2 and Figure 4). Neat PLA exhibited greater

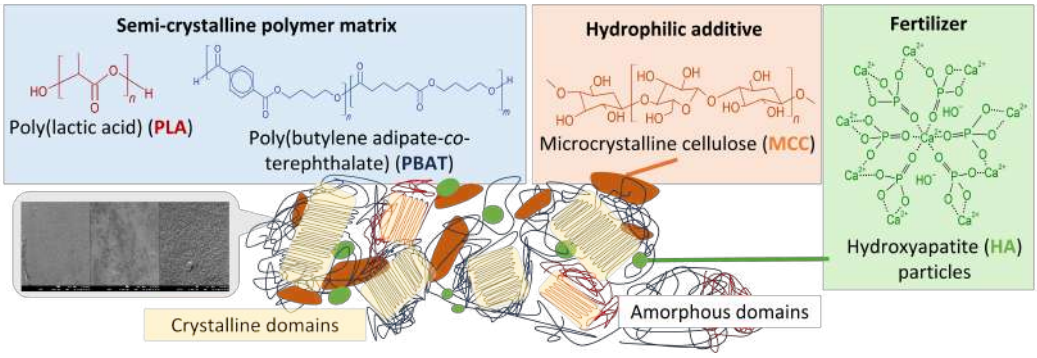


**Fig. 1** Diagram of constituents.

thermal resistance compared to neat PBAT, consistent with the relatively more stable aliphatic polyester backbone of PLA. Blended systems exhibited intermediate degradation behavior, indicating limited synergistic stabilization between the two polymer phases. In contrast, PBAT-rich formulations demonstrated earlier thermal degradation with higher char yields, attributable to the presence of thermally labile aliphatic adipate segments that undergo $\beta$-scission and chain cleavage at lower temperatures.

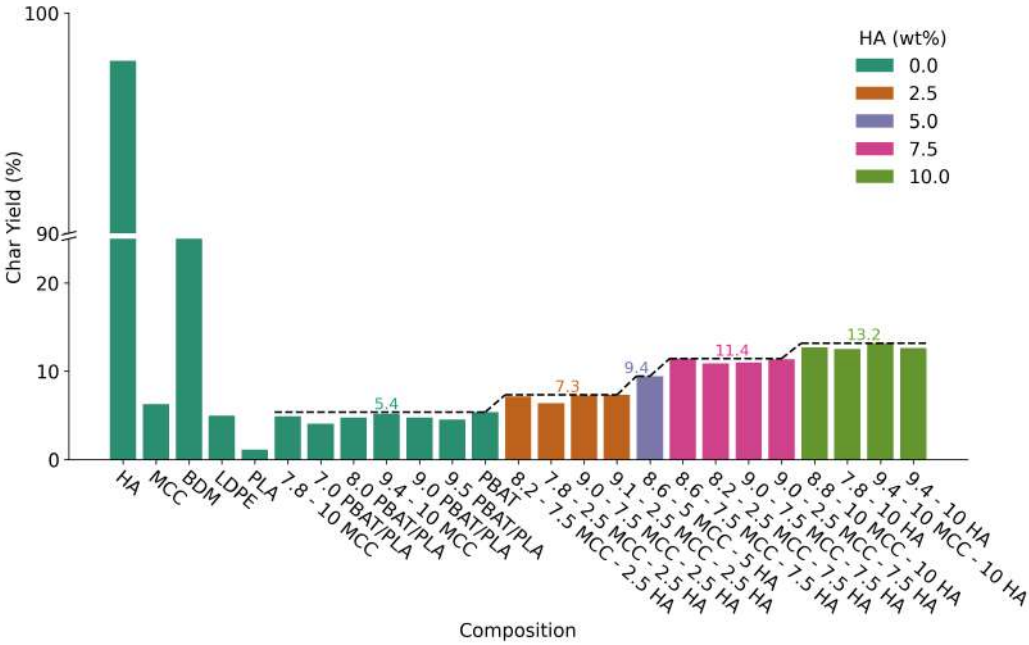


**Fig. 2** Char yield calculated for each composition using TGA traces.

Conversely, HA-containing formulations demonstrated increased residual mass at elevated temperatures. As an inorganic mineral phase, HA does not decompose under the tested conditions and contributes directly to char yield. In addition, HA particles may act as a physical barrier to heat and mass transfer, thereby promoting condensed-phase stabilization. Overall, thermal degradation behavior is governed by the balance between organic matrix decomposition and the stabilizing contribution of inorganic residue. TGA traces also revealed a large moisture content found in MCC at 4.7%, consistent with literature for a similar humidity as the locaiton of the TGA instrument [30]. This is interesting because although MCC was stored under nitrogen, its hydrophilic still allows for moisture retention during short periods of time of handling.

## 3.2 Water Absorption

The composite consists of a hydrophobic polymer backbone and particle additives (Figure 1). Water absorption was predominantly controlled by filler chemistry rather than matrix composition. Filler-free PLA/PBAT blends exhibited low equilibrium water uptake, ranging from approximately 0.34%

to 0.39%, regardless of PLA content. In contrast, systems containing 10 wt% MCC demonstrated substantially higher water absorption values (1.74-1.98%), corresponding to an approximately four- to six-fold increase relative to filler-free blends. Composite water absorptions ranged from 0.33% (20PLA-70PBAT-10HA) to 1.98% (20PLA-70PBAT-10MCC) (Figure 4).

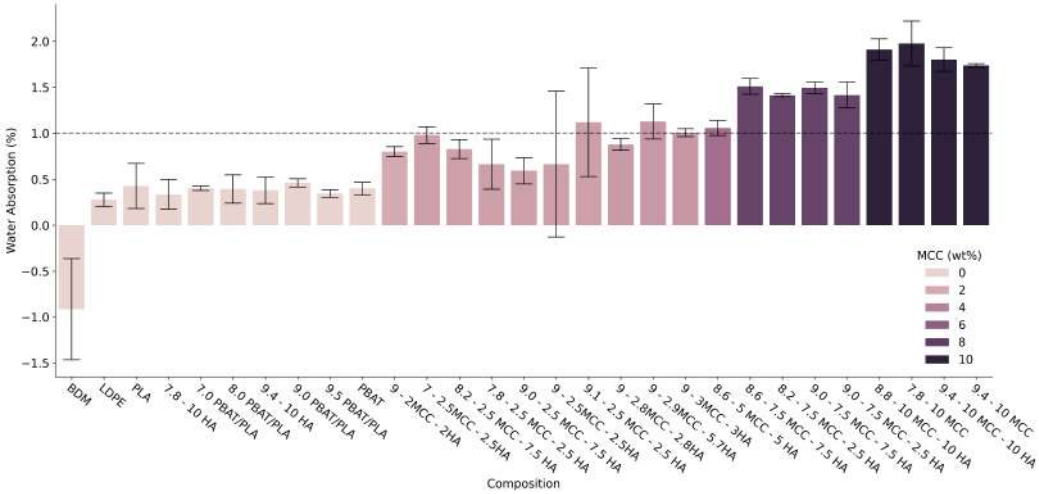


**Fig. 3** Average ± standard deviation (n=3) water absorption at saturation for each composition. The gray dashed line at 1% represents the typical upper limit for hydrophobic polymers.

Abiotic degradation is also a critical process in biodegradation. Water absorption results (Figure 3) show that the polyesters in the hydrophobic backbone do not contribute significantly to hydrophilic hydrogen bonding interactions and are in agreement with existing literature on water absorption of PLA/PBAT and LDPE [31]. HA does not exhibit significant hydrogen-bonding capacity or swelling behavior and therefore does not introduce comparable moisture diffusion pathways. On the contrary, the hydroxyl groups on the cellulose backbone act as both hydrogen bond donors and acceptors [12]. This hydrophobicity ¡ 1% characteristic of hydrophobic polymers is key to their durability as it prevents both hydrolysis and biodegradation [REFERENCE]. MCC had a strong impact on the water absorption (Figure 3 and Figure 4). Overall, hydrophilic interactions with MCC are expected due to its rich hydroxyl concentration. As degradation occurs, the interactions between cellulose and the hydrophobic polymer matrix are only expected to increase because the esters will be converted to carboxylic acid end groups during chain scission. Therefore, the polyesters will serve not only as hydrogen bond acceptors but hydrogen bond donors and increase hydrophilic interactions [12]. Sambudi et al. 2022 [32] saw a similar phenomenon with their composite made of orange peel powder and Ingeo$^{TM}$ 3251D PLA. The water absorptions of the neat PLA, PBAT, and LDPE were also consistent with literature. Water absorption is a critical parameter to measure because it will effect both abiotic and biotic BDM degradation [33]. Increasing polymer hydrophilicity can enhance biodegradation by promoting water uptake and accelerating hydrolytic chain scission, which lowers molecular weight and increases susceptibility to subsequent enzymatic/microbial assimilation [33]. This phenomenon is pH-dependent so the pH of the soil in which it is tilled will effect the swelling of the composite [32]. Water diffusion is highly dependent on polymer crystallinity, with amorphous regions being more permeable to water than crystalline domains [34]. Acidic (pH $\ll$ 7.4) or basic (pH $\gg$ 7.4) conditions are known to expedite hydrolytic degradation [34] via carbonyl protonation (and then $H_2O$ attack) or hydroxide attack, respectively [34]. Not only that, the swelling effect and mechanical stress induced by water retention can also facilitate biodegradation by allowing diffusion of enzymes [35]. We also expect sample dimensions to affect degradation in the future [34] – mulch films in practice will be thinner (typically 10 – 80 $\mu$m, but up to 150 $\mu$m for specialty crops) than the 300 $\mu$m thick samples we used in our experiments [13].

A correlation matrix, or heat map, was used as a preliminary assessment of multicollinearity or any correlations among factors (predictors) and outcomes (responses) (Figure 4). The heatmap is a representation of the Pearson correlation coefficient between each pair of features.

For example, Lendvai et al. [36] found that Ingeo$^{TM}$ 2003D PLA (∼4% D-isomer) had a water absorption of 0.71%. Ingeo$^{TM}$ 2003D and Ingeo$^{TM}$ 4032D (∼2% D-isomer) have similar molecular weights but differ in their D-isomer content [37]. The lower D-isomer content of Ingeo$^{TM}$ 4032D

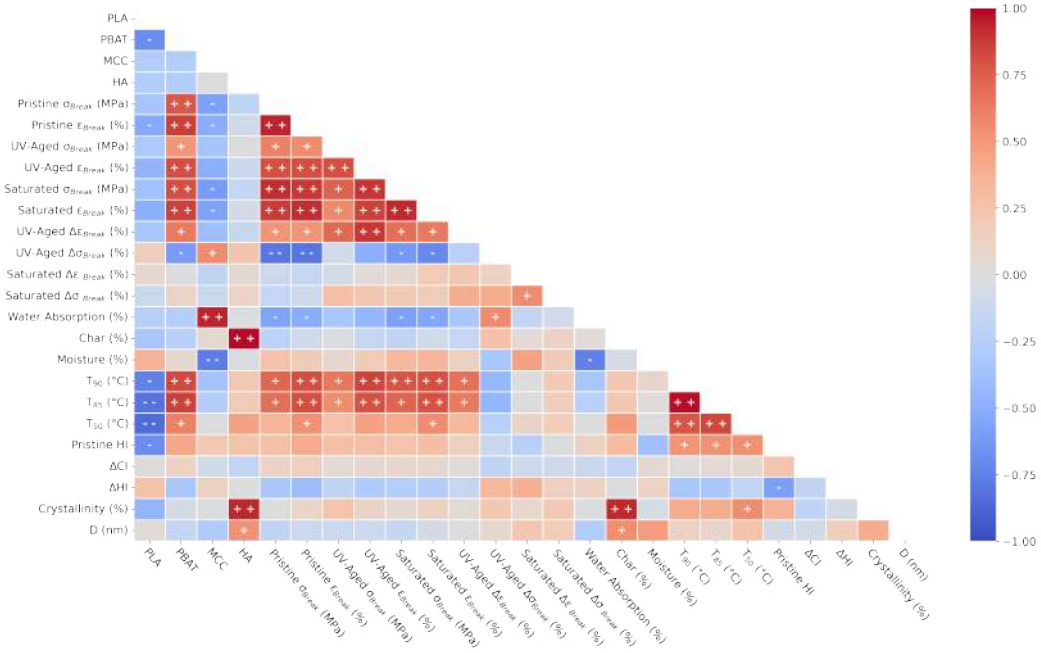


**Fig. 4** Heat map of correlation matrix for outputs where + + indicates r > 0.75, + indicates 0.5 < r ≤ 0.75, - indicates -0.75 < r ≤ -0.5, and - - indicates r < -0.75.

results in higher crystalline and mesophase concentrations compared to Ingeo$^{TM}$ 2003D [37], which would explain the lower water retention.

Moisture transport is not strongly morphology-dependent in this system. This suggests that water uptake is governed predominantly by the intrinsic hydrophilicity of the individual components rather than synergistic morphological effects. The BT blocks have a C:H:O ratio of 12:12:4 (co-block weight of 220.24 g·mol$^{-1}$) while the BA blocks have a ratio of 10:16:4 (co-block weight of 200.26 g·mol$^{-1}$). PLA has a C:H:O ratio of 3:4:2 and a monomer weight of 72.07 g·mol$^{-1}$. The overall shorter molecular weight of the PBAT polymer compared to the PLA would result in stronger interactions between cellulose due to end group interactions but this effect was still not dominant. PLA contains a higher concentration of oxygen (2,978 mole O · polymer$^{-1}$) than PBAT (697 mole O · polymer$^{-1}$), which leads to more prevalent interactions with MCC.

Interestingly, commercial BDM was the only composition to lose weight during water exposure (Figure 3). The composition of this BDM was not made available to us due to commercial propriety, but was advertised as containing various polyesters and starches. BDMs disintegrating in water has also been documented in literature by Tan et al. [38], who found that their poly(vinyl alcohol), kenaf fiber, and urea composite BDMs lost 25% of their initial weight after 7 days of submersion in water. This particular BDM also contained a large amount of additives, which could be aiding in disintegration, as evidenced by a char yield of 29% during TGA characterization (Figure 2).

## 3.3 Mechanical Testing

Mechanical properties are critical to application of agriculture mulch [31**?** ]. Neat PLA exhibited high tensile strength (∼71 MPa) but limited elongation at break (∼25%), confirming its brittle nature. Neat PBAT displayed lower tensile strength (∼45 MPa) but extremely high ductility (∼3300%). Mechanistically, PBAT imparts ductility, while PLA and rigid fillers restrict chain mobility, so elongation reflects the balance between brittle and ductile domains rather than a single dominant interaction. Interestingly, PLA concentration is typically strongly correlated with $\sigma_{\text{Break}}$ [31, 39], but in this design space was not.

An Elastic Net model was selected for regression because it was less sensitive to multi-collinearity than non-regularized linear models (e.g. ordinary least squares. Multi-collinearity was inherent in our design due to the mixture constraint of all values having to sum to 100%. Not only that, Elastic Net regression provides interpretable coefficients (where the coefficients provide mechanistic insight) and does not require parametric datasets for usage.

Blending PLA with PBAT resulted in intermediate mechanical behavior. Increasing PBAT content enhanced elongation while reducing tensile strength. For example, 5/95 PLA/PBAT exhibited ∼40 MPa tensile strength and ∼2616% elongation, whereas 20/80 PLA/PBAT exhibited ∼29 MPa tensile strength and ∼1912% elongation. The addition of 10 wt% MCC significantly reduced elongation (often below 500%), indicating stress concentration and restricted chain mobility. HA was less mechanically disruptive at equivalent loading.

Mechanical testing reflected a significant decrease in mechanical performance due to UV aging but not hydrolysis. However, the different durations of these periods and different temperatures during the exposure could be conflicting. For example, water absorption took place after 80 days because all samples were equilibrated with water prior to characterization while the UV aging duration was determined using Equation 2 and Equation **??**. UV exposure resulted in significant embrittlement across all compositions, which is consistent with literature [40]. The magnitude of ductility loss increased with PLA content, with 20/80 PLA/PBAT exhibiting approximately a 90% reduction in elongation after aging. Strength reductions were also observed, particularly in PBAT-rich systems. These changes are consistent with oxidative chain scission leading to reduced molecular weight and decreased entanglement density.

Vasylius et al. [40] evaluated the mechanical properties of LDPE after UV aging with a 2.45 $W \cdot m^{-2}$ UVA-351 fluorescence lamp for 1-40 hours, which is a higher intensity ( 80 $W \cdot m^{-2}$) at a shorter $\lambda$ (340 nm). Vasylius et al. [40] found that brittle-ness increased with as little as 1 hour of UV exposure. Interestingly, $\sigma_{\text{Break}}$ also increased with UV exposure [40].

Plasma degradation made the surface more hydrophilic [41]. They also saw a reduction in mechanical properties [41]. Observing the films after 3 months of submersion in soil revealed that plasma-treated films showed 1-3 um cracks while the not-treated sample did not show any. This treatment also improved cell adhesion of *Pseudomonas guariconesis*, which is a main factor in biofilm colonization [41].

Interestingly, in this small range of PLA concentrations that we chose (5–30%), PLA concentration weakly negatively correlated with stress at break. When the range is larger, PLA concentration strongly correlates. The mechanical properties of the films pristine but also as a function of aging was considered. For the application of mulch, it is useful to bifurcate degradation into two steps: abiotic degradation (when the film is applied on the soil surface and solar irradiation is intense) and biotic degradation (after tillage into the soil, where solar irradiation is diminished). The ideal composition is as durable as possible as a film but degrades as quickly as possible after tilled into particles. In this work, we characterize the abiotic degradation due to solar irradiation and water. Future work will consider biodegradation after tillage into soils as this is also a critical property.

MCC appeared to have a negative effect on mechanical properties (Figure **??**), which could be due to its relatively large particle size (on the order of $10^2$ $\mu$m) compared to HA (on the order of $10^1$ $\mu$m) (See Supporting Information. Particles can negatively impact the mechanical properties, as evidenced by work on PBAT/PLA/$CaCO_3$ system by Jia et al. 2024 [42]. CHAT GPT summary: They used solid-state shear-milling to disperse nano-$CaCO_3$ into PBAT/PLA and showed that good dispersion + in-situ compatibilization produced composites with improved tensile properties and balanced performance. The study highlights that well-dispersed nano-$CaCO_3$ at low wt% (typically 1–5 wt%) can improve properties, while poor dispersion drives deterioration.

Dil et al. 2016 [43] showed that the mechanical properties were enhanced with 3 wt% silica (fact check that). Scaffaro et al. 2020 did something with clay in PLA/PBAT composite [44] idk tbh. This paper, Ma et al. 2022 [45] is really good because it looks at the effects of the particles as a function of particle size on the mechanical properties.

Andrade et al. 2014 [46] characterize mulch films during three different stages: open field, tunnel, and open field. Mechanical properties were more complex and less predictable using composition as inputs than water absorption (Figure 6). Mechanical properties are highly dependent on processing and morphology which is not captured in composition [39]. In contrast, elongation at break exhibited moderate interaction dependence, suggesting that ductility arises from the compositional balance between brittle PLA, ductile PBAT, and rigid fillers. Ductility in multiphase polymer systems is governed by stress redistribution, phase continuity, and interfacial adhesion. In this system, PBAT contributes ductility, while PLA and rigid fillers restrict chain mobility. We expect the mechanical integrity and durability to be thickness-dependent, as was shown in LDPE by Xiong et al. [47].

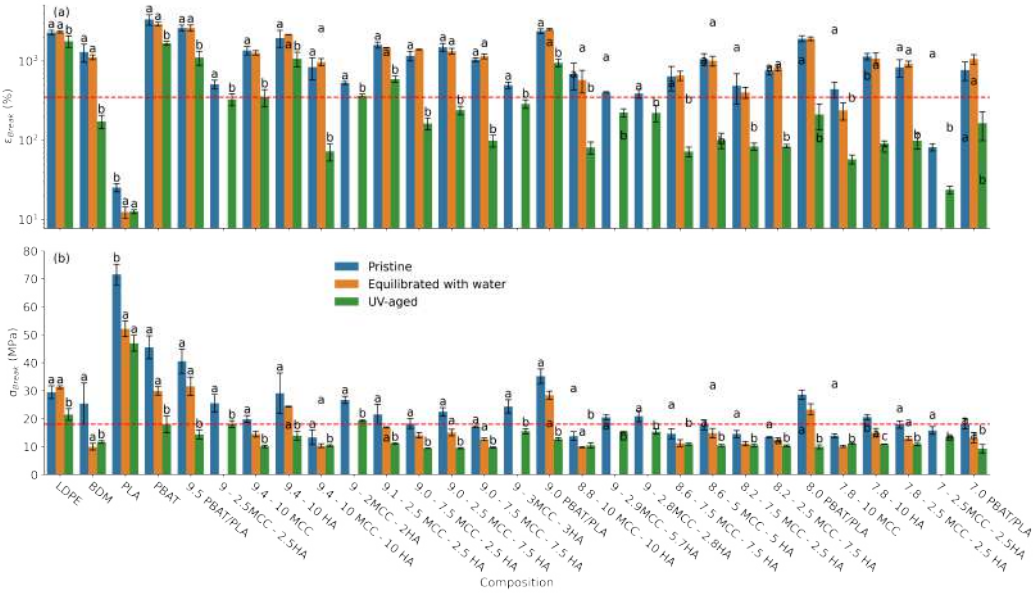


**Fig. 5** Average ± standard deviation (n=3) mechanical properties organized by composition for all runs. Mechanical properties include (a) $\varepsilon_{\text{Break}}$ and (b) $\sigma_{\text{Break}}$. Pair-wise significance results between pristine, UV-aged, and $H_2O$-saturated using Tukey's Honestly Significant Difference (HSD) test for each composition can be found in Supporting Information.

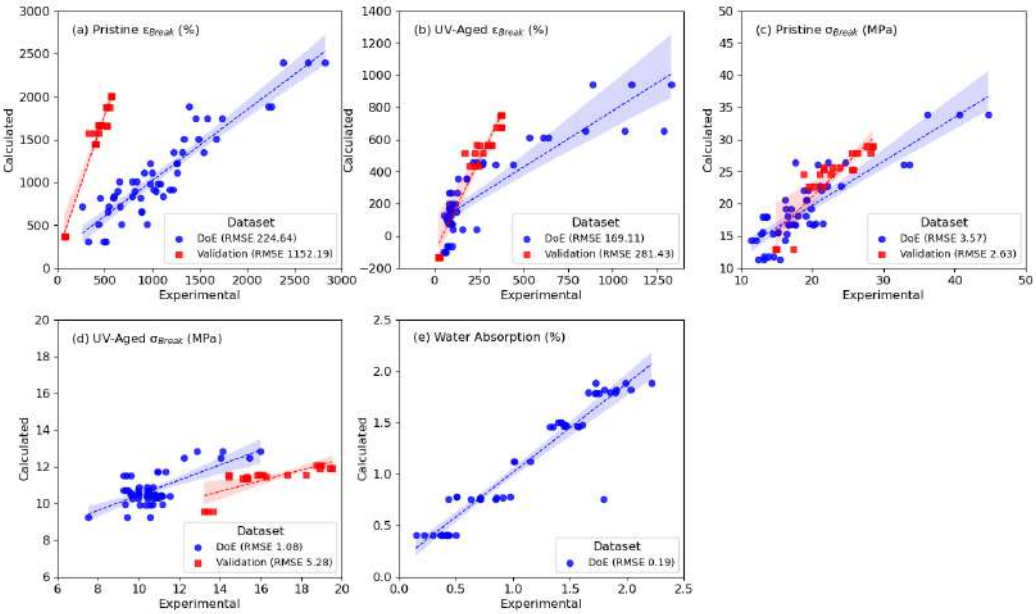


**Fig. 6** Predictions for (a) pristine $\varepsilon_{\text{Break}}$, (b) UV-aged $\varepsilon_{\text{Break}}$, (c) pristine $\sigma_{\text{Break}}$, (d) UV-aged $\sigma_{\text{Break}}$, and (e) $H_2O$ absorption. Shaded regions indicate 95% confidence bands.

## 3.4 Fourier-Transform Infrared Spectroscopy (FTIR)

FTIR analysis was conducted on pristine and UV-aged specimens to gain insight on polymer and additive chemistry, potential interactions, and aging progression. Graf et al. [11] evaluated the biodegradation of seven 10.5–29 $\mu$m thick BDMs made from PLA, PBAT, PHBV, and/or PAAU (and various additives) over 12 months in North Wales, U.K. after being applied to the soil or being buried as macroplastics. Irrespective of composition, BDMs showed larger changes in carbonyl index when buried as opposed to the abiotic damaged sustained while above ground [11]. Carbonyl index is used as an indicator of degradation because the ester bond is where cleavage occurs. PBAT-rich systems exhibited more pronounced CI increases, indicating greater susceptibility of aliphatic adipate segments to oxidative attack. However, all changes in CI between intra-composition pristine

**Table 2** Standardized coefficients for $\varepsilon_{\text{Break}}$, $\sigma_{\text{Break}}$, and Water Absorption.

| | Pristine | | | Aged | |
|---|---|---|---|---|---|
| Response | $\varepsilon_{\textbf{Break}}$ **(%)** | $\sigma_{\textbf{Break}}$ **(MPa)** | **Water Absorption (%)** | $\varepsilon_{\textbf{Break}}$ **(%)** | $\sigma_{\textbf{Break}}$ **(MPa)** |
| **PLA** | 0.426 | 0.482 | 0 | 0.286 | 0.870 |
| **PBAT** | 1.26 | 1.16 | 0 | 1.03 | 1.53 |
| **MCC** | 0 | 0 | 0.869 | 0 | 0 |
| **HA** | 0.346 | 0.196 | 0 | 0.757 | 0 |
| **PLA·PBAT** | 0 | 0 | 0 | 0 | 0 |
| **PLA·MCC** | 0 | 0 | 0.057 | 0 | 0.510 |
| **PBAT·MCC** | 0 | 0 | 0 | 0 | 0 |
| **PLA·HA** | 0.041 | 0.057 | 0 | 0 | 0 |
| **PBAT·HA** | 0 | 0 | 0 | 0.215 | 0 |
| **MCC·HA** | 0 | 0 | 0.0038 | 0 | 0 |

and aged samples were found to be statistically insignificant (alpha = 0.95). Weng et al. [48] tested PLA, PBAT, and a PLA/PBAT blend after 4 months of aging using FTIR.

Tao et al. [49] characterized FTIR for $CaCO_3$ with key peaks at 1391.24, 871.65, and 712.48 $cm^{-1}$.

Interestingly, for purposes of characterizing degradation, mechanical testing was more sensitive than FTIR. Bhattacharya et al. [41] reported significant decay of mechanical properties after plasma treatment but minimal changes with other characterization techniques (e.g. molecular weight, thermal properties). For example, the change in molecular weight via gel permeation chromatography after 10 s (from 51,000 $g{\cdot}mol^{-1}$ to 52,020 $g{\cdot}mol^{-1}$) and 20 s (from 51,000 to 50,400 $g{\cdot}mol^{-1}$) was likely statistically insignificant. Oxidation was perceptible using X-ray photon spectroscopy (XPS) [41]. This meant that the degradation was a surface, not bulk, phenomenon that did not dictate properties [41]. Similarly, it is possible that UV aging was limited to a surface that did not permit characterization via FTIR (micron-level penetration) but would be via XPS (nanometer-level penetration).

Carbonyl indices for polyolefins like PE and poly(propylene) found in literature range from 0-15, with researchers employing different methodologies (i.e. ranges used, FTIR equipment, and sample thicknesses) to determine the CI [50]. ATR is preferred over transmission FTIR because it is possible irrespective of sample thickness or embrittlement [50]. Reference for LDPE FTIR [51].

Xiao et al. 2025 [52] found that with UV aging, poly(styrene) exhibited carbonyl peaks. The presence of carbonyl peaks in our LDPE samples reflects their improper storage in a dark cabinet not under vacuum but under ambient conditions. Graf et al. 2025 [11] found that even pristine LDPE had some signs of carbonyl formation, with a starting CI of 0.33 [11]. Furthermore, their CIs ranged from 0.32-0.47 for LDPE, 0.91-2.36 for PLA/PBAT, and 16.03-31.51 for PBAT-based commercial BDMs [11]. The range for each varies. For example, Jubinville et al. 2023 [53] used the carbonyl range to be 1,850-1,650 $cm^{-1}$ with the $CH_2$ methylene region being from 1,500-1,420 $cm^{-1}$ while assessing the degradation of high-density poly(ethylene) (HDPE) during recycling. Xiong et al. 2023 [47] also found that "new" LDPE mulch contained carbonyl peaks associated with oxidation. The aging process is also affected by the thickness of the film. Thicker films yielded higher maize biomass and yields per land use, with the highest performing thickness being 0.015 mm (the largest film tested). They also quantified surface damage and found that: "The physical damage degree tended to increase with time, and followed a similar domino effect. Firstly, aging would cause dot-shaped damage, the number and area appeared to be relatively tiny. The tiny dot-shaped damages would further expand with the progress of aging, and be gradually connected up, forming larger area and number damages.

Interestingly, Xiong et al. [47] did not find a significant change in the carbonyl index before and after aging. The carbonyl index also did not depend on the thickness of the film. [47]. The carbonyl indexes for [47] ranged between $0.23 \pm 0.017$ (for a new 0.015 mm thick LDPE film) and $0.26 \pm 0.043$ (for an aged 0.006 mm thick LDPE film). The carbonyl indices found in this work instead ranged between 0.64 and 2.43. The hydroxyl indices found in this work ranged between 0.086 and 0.83. Xu et al. 2025 [54] show visible degradation in their films that is not present in the FTIR results. ¡– CHECK THIS CLAIM AND RE-READ THE PAPER

Carbery et al. 2023 [55] characterized the hydroxyl group index. During degradation of PA, HI increased by X% and Y% respectively after 12 months of aging [55].

While for our level of oxidation, FTIR was not suitable, future work will explore other characterization techniques like XPS which were useful for quantitatively analyzing oxidation of PE after plasma treatment [41]. The results that CI is not a sufficient method for quantifying degradation is supported by other researchers such as Hernandez-Charpak et al. [56], who found that CI did not

show any trends in aging via water submersion (hydrolysis), home composting, or industrial compositng of PBAT or PBAT/PLA blends. Instead, the viscosity when dissolved in chloroform and the cold crystallization temperature during cooling accurately correlated with the degree of aging, indicating a change in molecular weight. Meso-, macro-, and microfauna interactions with particles will also effect their degradation [11]. To evaluate this, Graf et al. [11] used a coarse and fine mesh system to allow for infiltration of mesofauna into the system. BDM degradation after tillage into soil is a primary concern for future work. BDMs had a negligible effect on the relative abundance of archaea, bacteria, and fungi which was interpreted as a sign that they are not being metabolized by the microbiota [11]. Bunsangiam et al. 2026 [57] characterized FTIR as a function of biotic aging.

### 3.5 X-Ray Diffraction (XRD)

XRD analysis revealed modest crystallinity across most experimental blends (approximately 3-7%), with most of the contribution of crystallinity originating from HA (Figure 7, Figure 4). The commercial BDM reference exhibited substantially higher crystallinity (∼45%), indicating a fundamentally different morphological structure compared to the experimental blends. Crystallite size variations were minimal across compositions, suggesting that differences in mechanical behavior are primarily governed by molecular weight changes and interfacial effects rather than substantial alterations in crystal dimensions.

XRD patterns revealed signature peaks for each constituent that allowed for the calculation of crystalline concentration. HA, as a highly crystalline inorganic material, contained a large array of peaks consistent with literature [58]. PLA, PBAT, MCC, and LDPE as semi-crystalline polymeric materials, contained less crystalline peaks with large amorphous halos (Figure 7, Supporting Information). PLA key peaks include 16.6 $\theta$. PBAT key peaks include 16.1, 17.5, 20.4, 23.3, and 25.1 $\theta$ [59]. MCC key peaks include 14.5, 16.9, 22.7, and 34.7 $\theta$ [60]. Key peaks for LDPE include 21.5, 23.9, 27.5, 29.5, 36.2, and 47.6 $2\theta$ [61].

Crystalline content was expected to be an indicator of degradation during UV aging. Literature supports that aging of polymers induces changes in crystalline structure due to the preferential degradation of amorphous regions [55]. For example, Carbery et al. [55] found that crystallinity via XRD decreased by 3-70% after 12 months of aging in a marine environment for LLDPE, PP, ePS, PET, PVC, PA, and PCL[55].

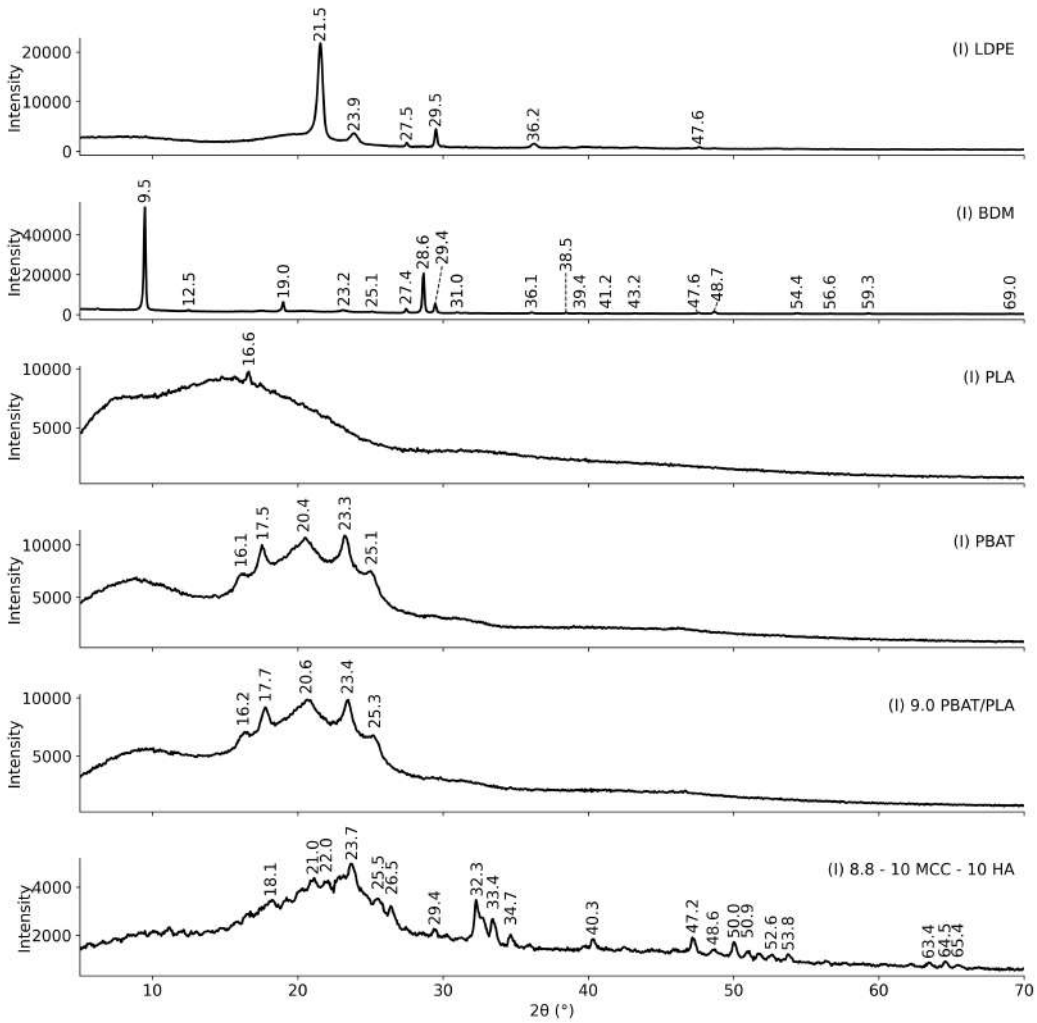


**Fig. 7** XRD diffractograms for each composition for pristine samples.

# 4 Conclusion

In conclusion, BDMs with eco-friendly additives were achieved using a PLA/PBAT matrix. PBAT concentration directly correlated with $\sigma_{\text{Break}}$ and $\varepsilon_{\text{Break}}$ but also with a larger reduction in mechanical properties due to UV-aging. MCC concentration directly correlated with $H_2O$ absorption but compromised pristine $\sigma_{\text{Break}}$ and $\varepsilon_{\text{Break}}$. The optimized composition for this work was 10.5% PLA, 83.8% PBAT, 2.8% MCC, and 2.8% HA. Decomposition due to UV-aging was not quantifiable using FTIR carbonyl index or SEM, but was detectable using tensile testing. Statistical evaluation confirmed a significant increase in water absorption with increasing MCC content and a significant reduction in elongation at break following UV exposure.

# Declarations

**Supplementary information.** Supporting information can be found....

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**Conflict of interest/Competing interests.** Conflict of interest/Competing interests

**Ethics approval and consent to participate.** Not applicable.

**Consent for publication.** Not applicable.

**Data availability.** Not applicable.

**Materials availability.** Not applicable.

**Code availability.** Not applicable.

**Author contribution.** Not applicable.

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# Supporting Information for PLA/PBAT/MCC/HA Composite Optimized for use as Biodegradable Mulch

DoE design parameters are listed in Table S1. Final hyperparameters used for ElasticNet regression are listed in Table S2. XRD Diffractograms for the full experimental set are found in Figure S1. FTIR traces for the full experimental set are found in Figure S8. Film topography and particle size of MCC and HA were characterized using JEOL JSM-7900F (JEOL USA Inc., Peabody, MA, USA) field emission scanning electron microscopy (SEM) and 10 nm 80Au-20Pd to minimize charging. Particle sizes were analyzed using Fiji ImageJ (Version 1.54p, National Institutes of Health, Bethesda, MD, USA) software. SEM images for the composite film and constituent particles (MCC and HA) can be found in Figures S2, S5, and S6, respectively. Additionally, the particle size distributions obtained from Figures S5, and S6, can be found in Figure S7. Note that the sample damage incured during imaging can be seen in Figure S2(g). An annotated version of the heat map with Pearson coefficients are included in Figure S9. DSC traces for MCC were obtained using a DSC 6000 (PerkinElmer Inc, Springfield, IL, USA) and 10-12 mg of sample in a crimp-sealed 50 $\mu$L vented alumninum pan, a heating rate of 10 °C·min$^{-1}$, and under 20 mL·min$^{-1}$ nitrogen flow for all temperature ramps. The temperature program followed: isotherm for 1.0 min at 30°C; heating from 30 to 200°C; isotherm for 3.0 min at 200°C; cooling from 200 to 0°C; and heating from 0 to 200°C. Figure S4 shows the results of DSC characterization, illustrating that MCC does not melt and instead undergoes significant thermal degradation during the first heating run. These results were also reproduced at 5 °C·min$^{-1}$. Full TGA and Derivative Thermogravimetry (DTG) traces are available in Figure S3.

**Table S1** DoE boundary conditions.

| Factor | Lower Limit | Upper Limit |
|---|---|---|
| PLA (wt%) | 5 | 30 |
| PBAT (wt%) | 70 | 95 |
| MCC (wt%) | 0 | 10 |
| HA (wt%) | 0 | 10 |

Source: The superscripts $^{D}$ and $^{C}$ represent the DoE and control datasets, respectively.

**Table S2** Tuned hyperparameters for Elastic Net regression.

| Response | $\alpha$ | L1 ratio |
|---|---|---|
| Pristine $\varepsilon_{\text{Break}}$ (%) | $10^{-6}$ | 1.0 |
| Pristine $\sigma_{\text{Break}}$ (MPa) | 0.00886 | 0.947 |
| Water Absorption (%) | 0.0298 | 1.0 |
| UV-Aged $\varepsilon_{\text{Break}}$ (%) | 0.0298 | 0.5 |
| UV-Aged $\sigma_{\text{Break}}$ (MPa) | $10^{-6}$ | 0.5 |

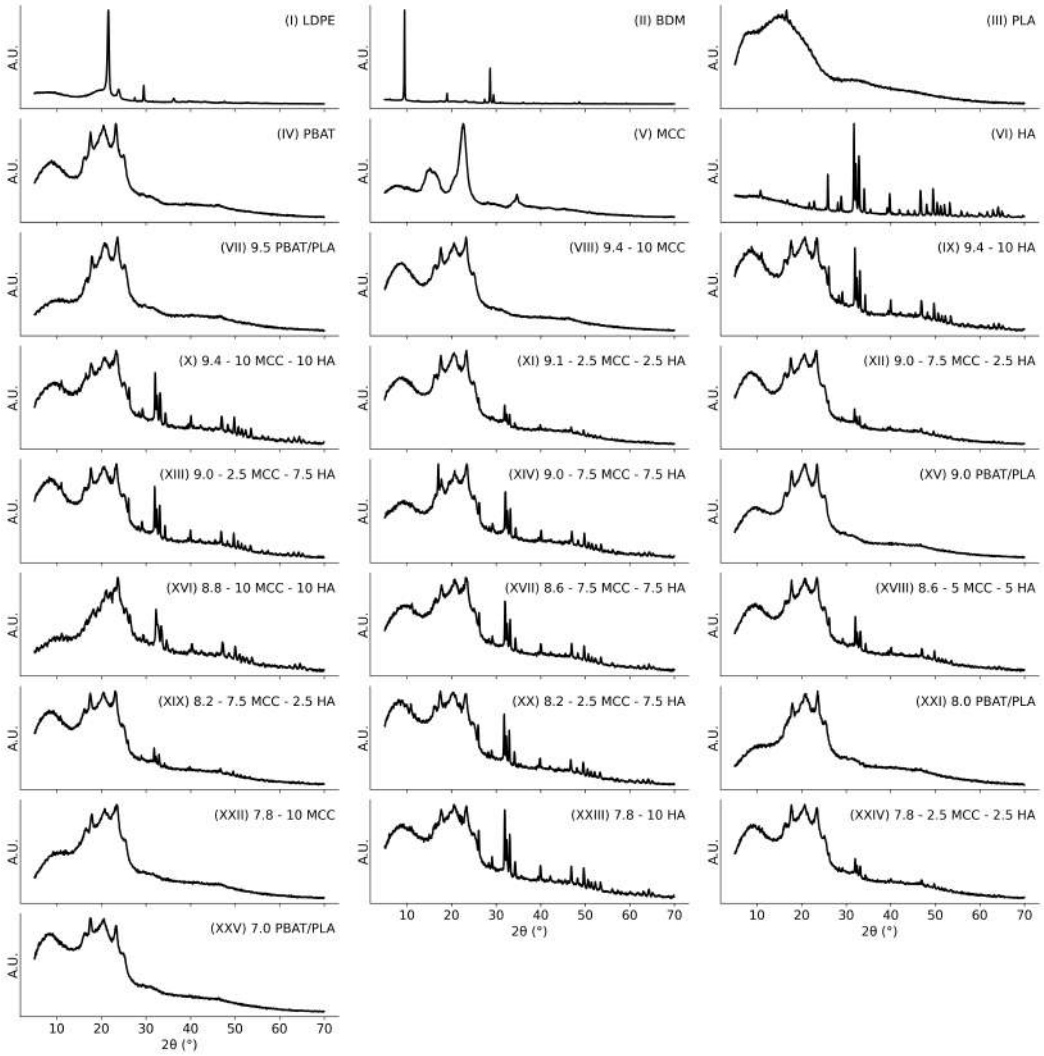


**Fig. S1** Full XRD traces for all compositions and control samples eith key peaks labeled.

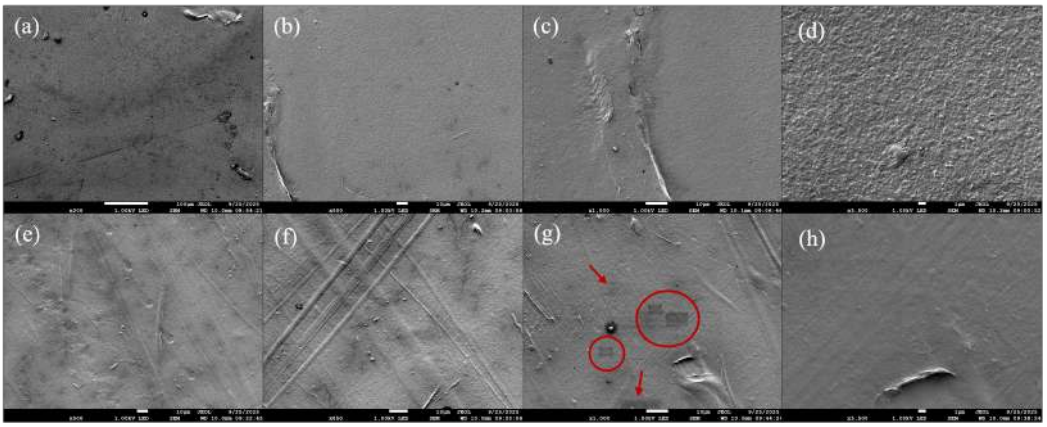


**Fig. S2** SEM images of various magnifications for pristine 8.75PLA-86.25PBAT-2.5MCC-2.5HA, taken as a representative sample for composite films. Red arrows indicate places where sample damage occured during scanning. Although lower voltages were attempted, a voltage which did not incur damage could not be performed and prevented us from doing additional characterization such as energy-dispersive X-ray spectroscopy

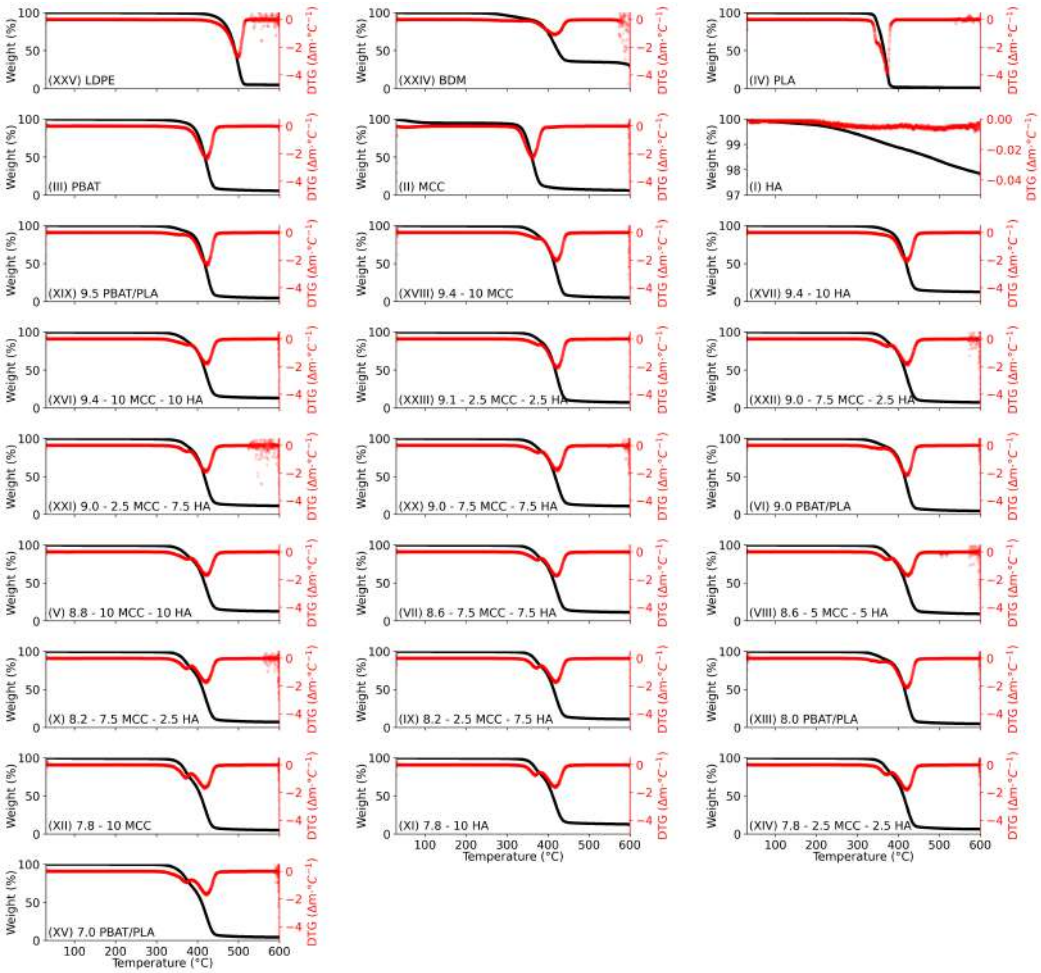


**Fig. S3** Full TGA (black) and DTG (red) traces for each composition.

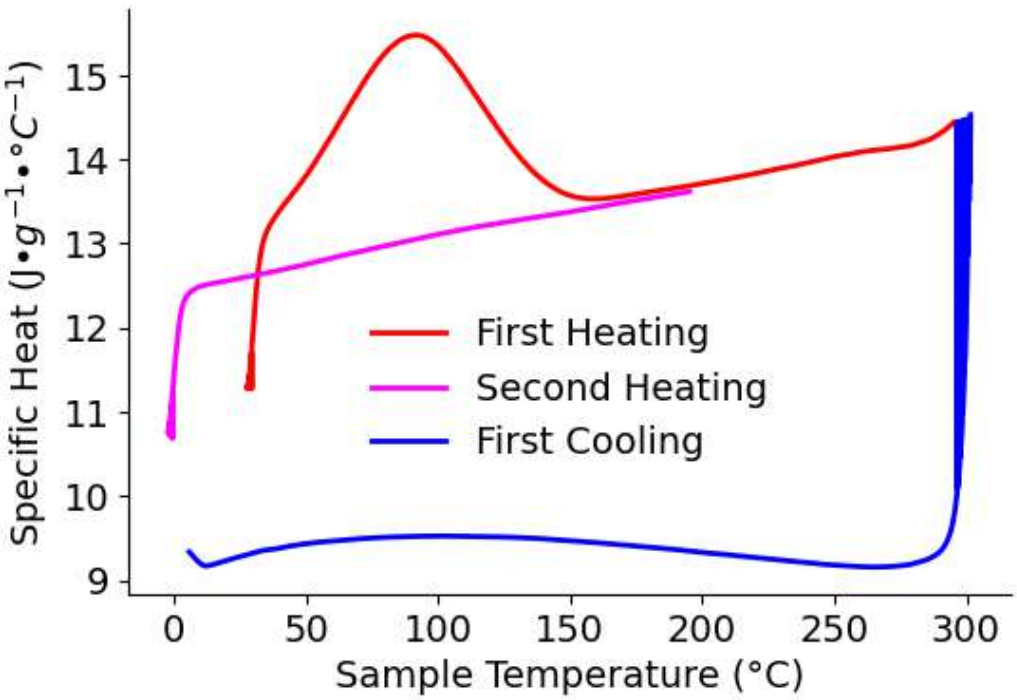


**Fig. S4** DSC thermogram (Endo Up) of commercial MCC powder.

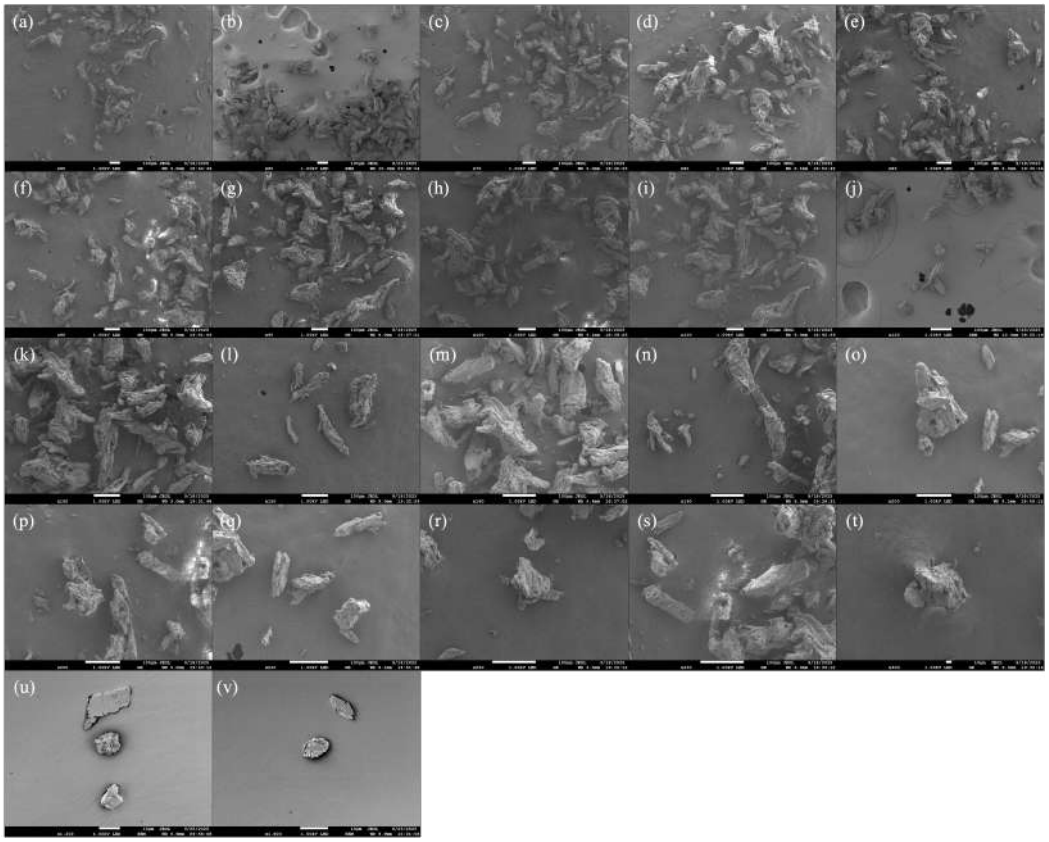


**Fig. S5** SEM images of various magnifications used for image processing for commercial MCC powder.

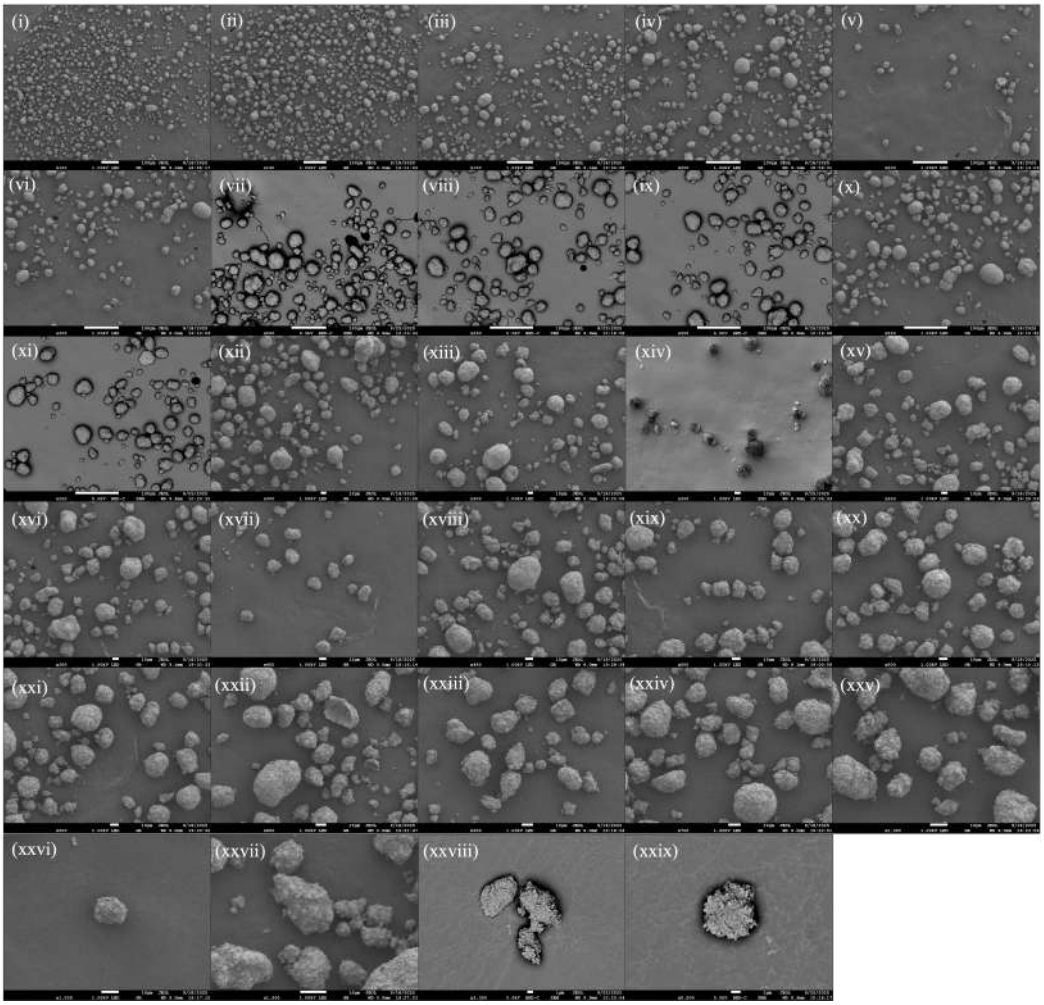


**Fig. S6** SEM images of various magnifications used for image processing for commercial MCC powder.

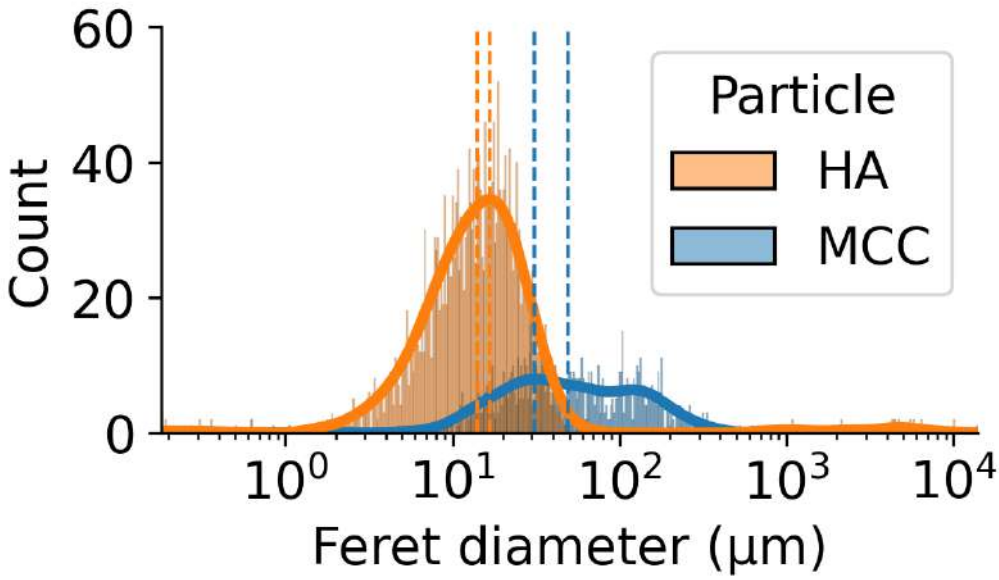


**Fig. S7** Particle size distribution of HA and MCC powders used in our compositions.The median and modes are indicated with orange (for HA) and blue (for MCC) dashed lines on the plot.

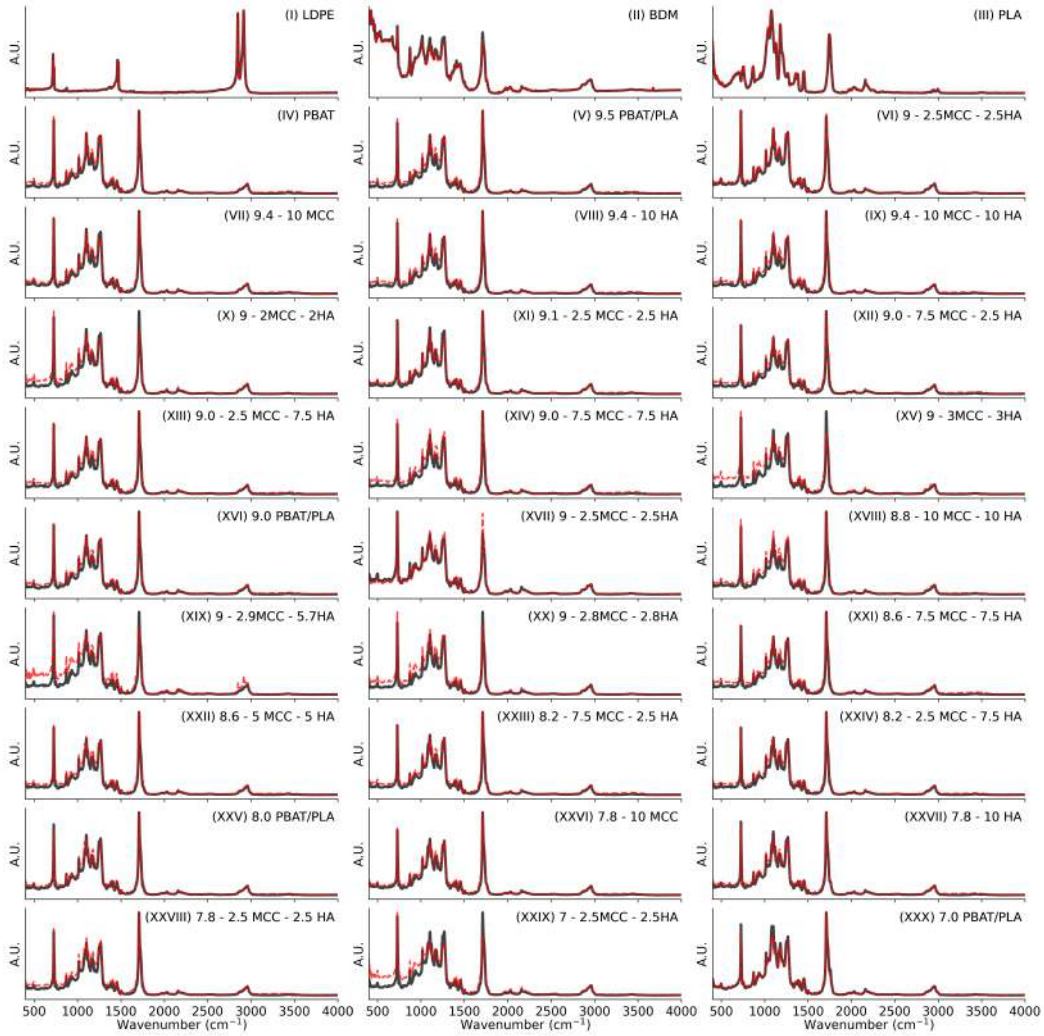


**Fig. S8** Representative FTIR traces from each sample before (solid lines) and after UV-accelerated aging (dashed lines) for each composition.

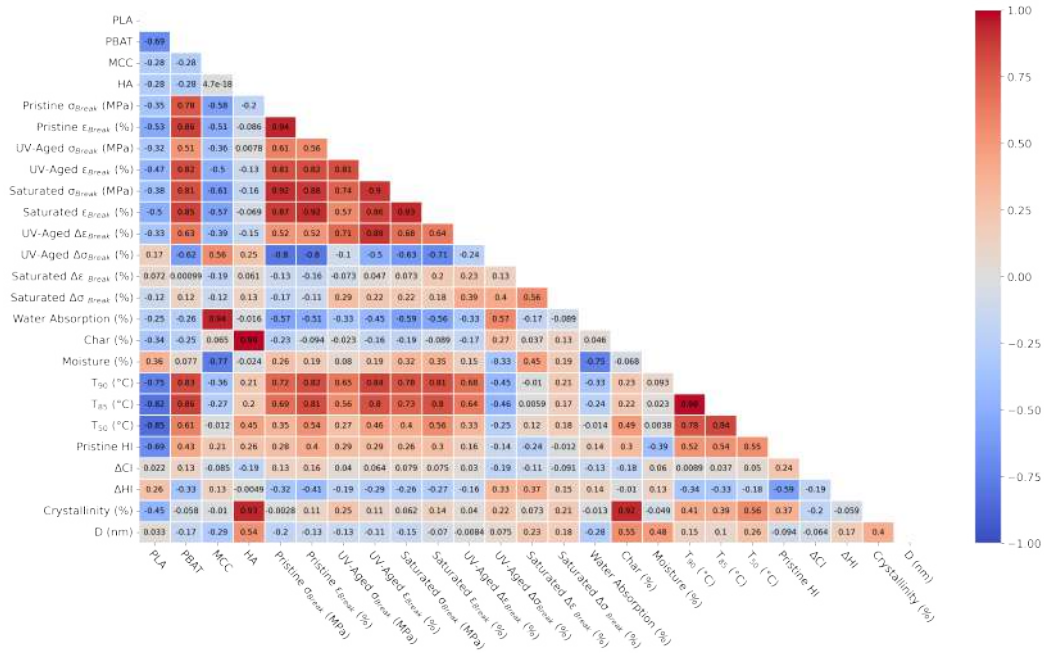


**Fig. S9** Heat map of correlation matrix for outputs with Pearson coefficients labeled.

**Table S3**: Results from characterization summarized.

| Run ID | PLA/PBAT/MCC/HA | Tensile Testing | | Water Absorption | TGA | | FTIR | | XRD | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | $\varepsilon_{\text{Break}}$ | $\sigma_{\text{Break}}$ | Water Absorption | Char Yield | Moisture | ΔCI | ΔHI | Crystallinity | D |
| | wt% | % | MPa | % | % | % | % | % | % | nm |
| **Design of Experiments (DoE)** | | | | | | | | | | |
| $1^{D}$ | 30 / 70 / 0 / 0 | 770.1 ± 202.63 | 18.1 ± 1.79 | 0.4 ± 0.02 | 4.05 | 0.0 | 9.64 ± 20.44 | 6.38 ± 2.32 | 2.81 | 95.3 |
| $2^{D}$ | 21.25 / 73.75 / 2.5 / 2.5 | 831.5 ± 203.72 | 17.8 ± 1.32 | 0.66 ± 0.27 | 6.36 | 0.2 | 3.44 ± 5.19 | 0.08 ± 3.24 | 3.43 | 113.7 |
| $3^{D}$ | 20 / 70 / 10 / 0 | 473.3 ± 39.59 | 13.6 ± 1.05 | 1.74 ± 0.03 | 7.92 | 0.4 | 2.53 ± 1.67 | 6.97 ± 10.3 | 3.01 | 53.4 |
| $4^{D}$ | 20 / 70 / 0 / 10 | 1240.4 ± 311.05 | 19.4 ± 1.81 | 0.5 ± 0.07 | 6.47 | 0.1 | 2.26 ± 0.95 | -6.36 ± 5.74 | 3.34 | 151.4 |
| $5^{D}$ | 16.25 / 73.75 / 7.5 / 2.5 | 613.5 ± 131.36 | 16.7 ± 2.81 | 1.13 ± 0.04 | 9.78 | 0.5 | 13.52 ± 1.96 | -2.73 ± 2.7 | 5.74 | 136.0 |
| $6^{D}$ | 16.25 / 73.75 / 2.5 / 7.5 | 757.8 ± 70.1 | 18.1 ± 2.67 | 1.39 ± 0.02 | 9.53 | 0.3 | 0.19 ± 0.69 | 4.34 ± 2.36 | 5.13 | 166.1 |
| $7^{D}$ | 12.5 / 77.5 / 5 / 5 | 1184.8 ± 263.43 | 21.5 ± 2.25 | 1.06 ± 0.01 | 10.24 | 0.3 | 4.53 ± 3.4 | 4.37 ± 6.25 | 4.96 | 86.2 |
| $8^{D}$ | 11.25 / 73.75 / 7.5 / 7.5 | 553.8 ± 109.8 | 16.1 ± 3.51 | 1.73 ± 0.04 | 12.24 | 0.3 | 4.08 ± 1.44 | 3.3 ± 3.74 | 7.25 | 158.7 |
| $9^{D}$ | 10 / 70 / 10 / 10 | 482.7 ± 38.45 | 15.8 ± 1.15 | 2.2 ± 0.02 | 13.11 | 0.6 | 1.64 ± 1.64 | -1.0 ± 2.42 | 5.5 | 71.9 |
| $10^{D}$ | 8.75 / 81.25 / 7.5 / 2.5 | 1164.1 ± 164.98 | 18.2 ± 1.76 | 1.42 ± 0.14 | 7.27 | 0.2 | 3.7 ± 5.72 | -3.86 ± 3.84 | 3.88 | 105.5 |
| $11^{D}$ | 8.75 / 86.25 / 2.5 / 2.5 | 1593.8 ± 139.47 | 21.5 ± 3.57 | 1.1 ± 0.59 | 7.3 | 0.2 | 11.6 ± 15.36 | 0.7 ± 0.58 | 4.0 | 139.8 |
| $12^{D}$ | 8.75 / 81.25 / 2.5 / 7.5 | 1123.3 ± 243.08 | 18.0 ± 2.19 | 1.4 ± 0.04 | 10.3 | 0.3 | 5.1 ± 9.75 | -6.6 ± 7.77 | 5.5 | 146.5 |
| $13^{D}$ | 8.75 / 76.25 / 7.5 / 7.5 | 530.5 ± 156.27 | 14.3 ± 1.91 | 2.1 ± 0.08 | 12.3 | 0.5 | -0.4 ± 3.23 | -6.8 ± 6.77 | 7.6 | 145.7 |
| $14^{D}$ | 5 / 75 / 10 / 10 | 837.1 ± 259.99 | 13.3 ± 2.61 | 1.8 ± 0.13 | 13.17 | 0.5 | 0.87 ± 0.84 | -2.68 ± 3.58 | 7.02 | 129.1 |
| $15^{D}$ | 5 / 95 / 0 / 0 | 2616.0 ± 222.33 | 40.5 ± 4.3 | 0.3 ± 0.04 | 4.53 | 0.2 | 6.0 ± 9.91 | 1.34 ± 1.37 | 4.29 | 30.5 |
| $16^{D}$ | 5 / 85 / 0 / 10 | 1954.9 ± 490.22 | 29.1 ± 7.24 | 0.38 ± 0.14 | 12.61 | 0.1 | 1.68 ± 1.02 | -8.81 ± 6.75 | 6.86 | 158.5 |
| $17^{D}$ | 5 / 85 / 10 / 0 | 1362.3 ± 163.88 | 19.8 ± 1.26 | 1.74 ± 0.02 | 5.18 | 0.4 | 8.75 ± 6.97 | -2.65 ± 5.3 | 3.22 | 35.2 |
| **Control Samples** | | | | | | | | | | |
| LDPE | N/A | 2295.7 ± 152.77 | 29.5 ± 2.29 | 0.3 ± 0.09 | 5.0 | 0.0 | 19.6 ± 2.21 | -3.5 ± 7.15 | 21.9 | 119.5 |
| BDM | N/A | 1298.0 ± 332.53 | 25.4 ± 7.36 | -0.9 ± 0.55 | 29.0 | 0.2 | 2.6 ± 2.88 | -0.3 ± 4.4 | 45.7 | 209.0 |
| $18^{C}$ | 100 / 0 / 0 / 0 | 2023.3 ± 418.69 | 61.9 ± 3.19 | 0.3 ± 0.07 | 0.0 | 0.1 | -0.92 ± 5.97 | -2.74 ± 7.81 | 20.36 | 106.6 |
| $19^{C}$ | 0 / 100 / 0 / 0 | 3338.3 ± 504.5 | 45.5 ± 4.1 | 0.4 ± 0.07 | 5.4 | 0.3 | -4.7 ± 3.48 | -5.2 ± 5.23 | 5.03 | 36.2 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| $20^{C}$ | 0 / 0 / 100 / 0 | N/A | N/A | 0.4 N/A | CHAR | MOISTURE | N/A | N/A | x | D |
| $21^{C}$ | 0 / 0 / 0 / 100 | N/A | N/A | 0.4 N/A | CHAR | MOISTURE | N/A | N/A | x | D |
| **Validation Samples** | | | | | | | | | | |
| $22^{V}$ | 10.5 / 83.8 / 2.8 / 2.8 | 523.2 ± 71.33 | 26.0 ± 1.98 | 0.97 ± 0.04 | – | – | 15.41 ± 1.64 | 1.99 ± 2.84 | – | – |
| $23^{V}$ | 10.3 / 81.1 / 2.9 / 5.7 | 371.1 ± 12.09 | 23.0 ± 1.54 | 1.41 ± 0.05 | – | – | 3.41 ± 4.3 | -4.77 ± 8.14 | – | – |
| $24^{V}$ | 5 / 90 / 2.5 / 2.5 | 492.1 ± 56.79 | 22.1 ± 1.58 | 0.97 ± 0.02 | – | – | 3.92 ± 10.54 | 1.97 ± 10.48 | – | – |
| $25^{V}$ | 8 / 88 / 2 / 2 | 569.5 ± 25.1 | 24.9 ± 0.16 | 1.01 ± 0.01 | – | – | 3.71 ± 8.15 | -1.45 ± 4.35 | – | – |
| $26^{V}$ | 9 / 85 / 3 / 3 | 416.2 ± 46.79 | 25.5 ± 0.23 | 1.09 | – | – | 10.05 ± 1.27 | 0.93 ± 3.84 | – | – |
| $27^{V}$ | 10 / 85 / 2.5 / 2.5 | 451.6 ± 18.05 | 22.7 ± 1.14 | 0.99 ± 0.01 | – | – | 10.47 ± 10.63 | 0.07 ± 10.55 | – | – |
| $28^{V}$ | 29 / 66 / 2.5 / 2.5 | 81.9 ± 8.68 | 15.8 ± 1.39 | 0.98 ± 0.09 | – | – | -1.44 ± 9.91 | -1.44 ± 5.51 | – | – |

Moisture (%) is determined from TGA. Mechanical properties reported here are for pristine samples. Superscripts $^{D}$, $^{C}$, and $^{V}$ represent the DoE, control, and validation datasets, respectively.

Oxygen concentrations for PLA and PBAT were calculated to predict interractions between constituents based on hydrogen bonding. An Mn of 107,296 g·mol$^{-1}$ [62] for Ingeo$^{TM}$ 4032D and an Mn of 36,500 for PBAT C1200 g·mol$^{-1}$ from Bojda et al. [63] were assumed. PLA polymer molar oxygen concentration was calculated using:

$$N_{\text{O, PLA}} = \frac{107,296\ g \cdot mole\ polymer^{-1}}{72.07\ g \cdot mole\ monomer^{-1}} \cdot \frac{2\ mole\ O}{1\ mole\ monomer} = 2,978\ mole\ O \cdot\ mole\ polymer^{-1}$$

For PBAT, the concentration of oxygen was calculated separately for each co-block and then added together. Concentrations of 51.6% BA and 48.4% BT from Wojtcak et al. 2014 [59] using H-NMR) were assumed. The BA segment oxygen concentration was calculated using:

$$N_{\text{O, PBAT, BA}} = \frac{0.516 \cdot 36,500\ g \cdot mole\ polymer^{-1}}{200.26\ g \cdot mole\ BA\ monomer^{-1}} \cdot \frac{4\ mole\ O}{1\ mole\ BA\ monomer} = 376\ mole\ O \cdot mole\ polymer^{-1}$$

and then for the BT segment:

$$N_{\text{O, PBAT, BT}} = \frac{0.484 \cdot 36,500\ g \cdot mole\ polymer^{-1}}{220.24\ g \cdot mole\ BT\ monomer^{-1}} \cdot \frac{4\ mole\ O}{1\ mole\ BT\ monomer} = 321\ mole\ O \cdot mole\ polymer^{-1}$$

In total, this yields 697 moles of oxygen per mole of polymer.

UV aging studies were done in "batches" due to space and sample preparation constraints. The target exposure was calculated using:

$$(1000\ kW \cdot hour \cdot m^{-2} \cdot year^{-1}) \cdot (0.05) \cdot \frac{3\ months}{12\ months \cdot year^{-1}} = 12.5\ kW \cdot hour \cdot m^{-2}$$

Irradiation time for each batch of UV aging was as follows:

$$Batch\ 1: \frac{12.5\ kW \cdot hour \cdot m^{-2}}{0.0806394\ kW \cdot m^{-2}} \cdot \frac{1\ day}{16\ hours\ LIGHT\ ON} = 9.688\ days$$

$$Batch\ 2: \frac{12.5\ kW \cdot hour \cdot m^{-2}}{0.095638\ kW \cdot m^{-2}} \cdot \frac{1\ day}{16\ hours\ LIGHT\ ON} = 8.169\ days$$

$$Batch\ 3: \frac{12.5\ kW \cdot hour \cdot m^{-2}}{0.091861\ kW \cdot m^{-2}} \cdot \frac{1\ day}{16\ hours\ LIGHT\ ON} = 8.505\ days$$

The figures where irradiation was measured and the on/off cycles for temperature can be found in Figures S10 and S11, respectively.

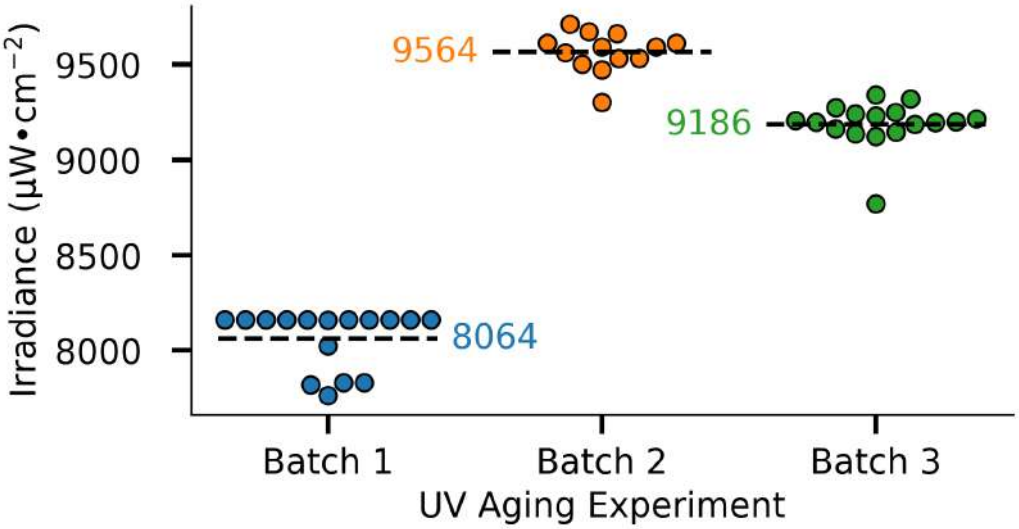


**Fig. S10** Irradiation profiles for each batch of UV aging experiments.

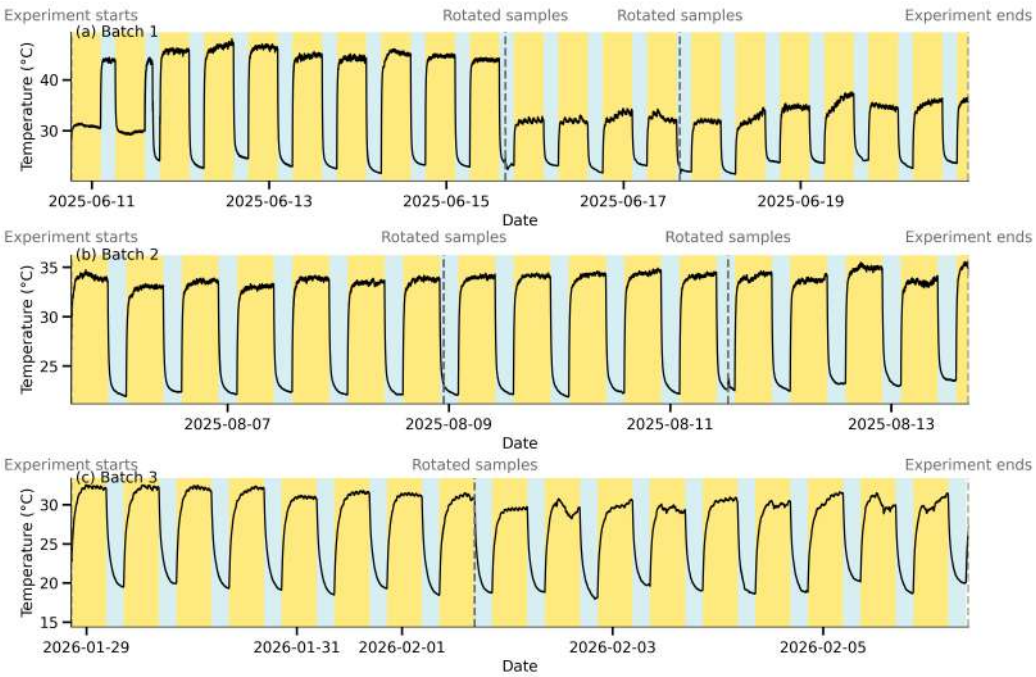


**Fig. S11** Temperature profiles and cycles for each batch of UV aging experiments. Note that in Batch 1 the temperature sensor was moved during sample rotation and there was a device power issue that changed the temperature profile briefly for the first day.